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Effect of age on anticipatory postural adjustments in unilateral arm movement

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Abstract

Aging is frequently accompanied by a deterioration in postural control. Accordingly, the elderly adopt postural strategies in order to maintain balance. The purpose of this study was to compare anticipatory postural adjustments in (healthy) 10 young and 10 elderly subjects using electromyography (EMG) and biomechanical parameters. While standing on a force platform, subjects performed voluntary, arm-raising movements under five conditions: self-paced at three different velocities, self-paced with load and an externally triggered, both at maximal velocity. The force platform provided information on vertical torque (T_z) and center of pressure anteroposterior displacements (COP). EMG activity was recorded from the biceps femoris, quadriceps, tibialis anterior and soleus muscles. Voluntary movements were associated with an early COP backward shift and an anticipatory T_z. At low velocity, elderly subjects did not show any impairment in stability. At maximal velocity, T_z was delayed in all conditions in the elderly group, whereas COP latency was reduced only in the self-paced condition without load. Despite this decrease in anticipation, the movement was performed at the same velocity as in younger subjects. The elderly adopted various muscle strategies in order to perform the same movement with less stability. In the self-paced condition, elderly subjects used a hip strategy, whereas young subjects used an ankle strategy. In the triggered condition, the strategy corresponded to increased activation of certain thigh muscles, rather than a sequence modification. Hence, local muscle strategies were used to counteract the overall delay in postural preparation revealed by biomechanical parameters.
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1. Introduction

Execution of a voluntary movement (such as arm movement performed when standing) is preceded by a sequence of postural modifications which are reflected by recruitment of the postural (leg and trunk) muscles associated with accelerations of the various body segments [1–3]. These anticipatory postural adjustments (APAs) are movement-specific, and thus, appear to be centrally organized prior to movement execution [4]: they tend to minimize the potential disturbance of the forthcoming movement and allow equilibrium to be maintained during execution.

The postural control system has been shown to decline with age. Elderly subjects often experience a deterioration in the integrity of many of the physiological systems which participate in the control of postural stability [5–9]. As the quality of sensory information decreases, postural stability control becomes increasingly difficult and requires a wider safety margin [10]. Many authors have used reaction time conditions to study the postural strategies adopted by the elderly when they perform upper limb movements [11,12]. In these studies, postural adjustments were analyzed through various parameters, such as center of pressure (COP) displacement or postural and voluntary muscle activities.

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During pull-and-push arm movements, the elderly show deficiencies or slowness in achieving a safe degree of postural stability prior to initiation of voluntary movement [10]. By studying the same movement paradigm in response to a signal during sitting or in the standing position, Stelmach et al. [13] showed that elderly subjects reacted more slowly in the standing condition and concluded that the slowness was associated with a stabilization deficiency. Inglin and Woollacott [11] observed a delayed latency in elderly subjects for both postural and voluntary muscles, with a greater time interval between activation of these two muscle groups. In contrast, other authors studying postural adjustments in elderly subjects (unilateral shoulder flexion at different velocities) have noted that delays in postural muscle activation resulted in decreased latency between postural and voluntary muscle activation [12,14]. The authors observed age-related differences (notably delayed activation) in the timing of two postural muscles and concluded that the elderly may not be as proficient in controlling movement under very fast or very slow conditions [12,14]. For an arm-raising movement, a wider range of values for COP excursion speed was observed for elderly subjects, compared with young subjects [15].

In view of the discordance between these various results, we thus decided to study the control of posture associated with a voluntary, unilateral arm flexion performed by young and elderly subjects in the standing position. We investigated APAs evidenced by biomechanical parameters (such as COP movements and vertical torque (T_z)) and by EMG data during different movement paradigms, in order to compare age-related differences in postural adjustment. In a previous study, we had demonstrated that vertical torque (T_z) is a useful parameter for quantifying APA associated with unilateral movement [16,17].

The goal of this research is to provide additional information on the organization of postural control in elderly subjects performing arm raising, a common everyday movement. A number of questions were addressed: are APAs altered under all types of movement conditions? And if so, which strategies are used to maintain posture and equilibrium?

2. Methods

2.1. Subjects

We examined 20 healthy, right-handed adult volunteers with no history of neurological or motor disorders. Subjects were classified into two age groups consisting of 10 individuals each. The mean age of the young group (which included five males and five females) was 29.4 ± 8.4 years (±S.D.), whereas that of the elderly group (eight males and two females) was 67.3 ± 4.8 years. The mean body mass and height for the younger subjects were 66.8 ± 11.5 kg and 169.6 ± 10.3 cm, respectively, for the elderly subjects, these values were 74.4 ± 9.5 kg and 170.1 ± 6.5 cm, respectively. All subjects gave their written, informed consent to participation in the study, which was approved by the local ethics committee.

2.2. Procedure

Subjects (barefoot) stood upright on a force platform (AMTI OR-6, Advanced Mechanical Technology Inc., Newton, MA, USA) with equal weight on each foot, looking ahead and with their arms hanging freely alongside the body. During all trials, foot position remained the same. Subjects were asked to perform shoulder flexion (with elbow extension) with their dominant arm in order to reach and grasp a handle located in front of them. This handle was placed so as to be within the subject's grasp when the arm was held at 90° relative to the bodyline. Movements were rejected if they were accomplished with elbow flexion. The focal movement was executed under different conditions:

(1) A self-paced task at three different velocities (low, medium and maximal). Subjects performed the movement at any self-selected time following a starting cue given by the experimenter. They were thus free to choose the instant of movement onset.
(2) A reaction time task at maximal velocity (without load), in which subjects initiated the movement as soon as possible after a sound signal.
(3) A self-paced task at maximal velocity, with a load of 1 kg attached to the wrist.

The subjects were encouraged to make all the movements in all conditions as similar to each other as possible. The velocity was verified after each trial: if subjects made very rapid or very slow movements, they were asked to slow down or speed up their movements accordingly. For each condition, subjects briefly maintained the final position before lowering the arm. After a brief practice period, each subject performed 10 trials for each of the different movement conditions (i.e. 10 trials × 5 conditions). The order of condition was the next: self-paced in increasing velocity, triggered and loaded.

2.3. Recording system and data analysis

The movement was recorded using a VICON 370 optoelectronic system (Oxford Metrics Ltd., Oxford, UK) with six infrared video cameras and 16 retro-reflective, spherical (25 mm diameter) markers fixed on several anatomical sites: the spinous process of the seventh cervical vertebra, the sacrum and bilaterally on the acromion, the lateral epicondyl of the elbow, the radial styloid of the wrist, the anterior superior iliac spine, the lateral tuberosity of the knee, the external malleolus of the ankle and the base of the second toe. The markers on the right acromion and wrist were used to compute the arm's angular displacement and

angular velocity (i.e. the first derivative of the displacement) (Fig. 1). The force plate signals (sampled at 250 Hz) provided the three components of the floor reaction forces (F) in the anteroposterior (x), lateral (y) and vertical (z) directions, as well as the three components of the moment (M). We used these data to compute the coordinates of the centre of pressure (CoP) in both anteroposterior (x_{CoP}) and lateral directions (y_{CoP}). The vertical torque (T_z) was also calculated, using the equation: $T_z = M_z + y\text{COP} \times F_x - x\text{COP} \times F_y$, where the term ($y\text{COP} \times F_x - x\text{COP} \times F_y$) is used to take into account the variable distance from the centre of the force plate [2,16,18].

Kinetic and kinematic data were recorded simultaneously. Kinematic data were recorded at 50 Hz and then linearly interpolated at 250 Hz, to allow synchronization with analog data. EMG activity was recorded using bipolar surface electrodes (Motion Control Inc., Utah, USA) placed over the bellies of the right anterior deltoid and (bilaterally) the biceps femoris (BF), quadriceps, tibialis anterior (TA) and soleus muscles (Fig. 2). Signals were band-pass filtered (20–150 Hz), rectified, amplified and then sampled at 250 Hz.

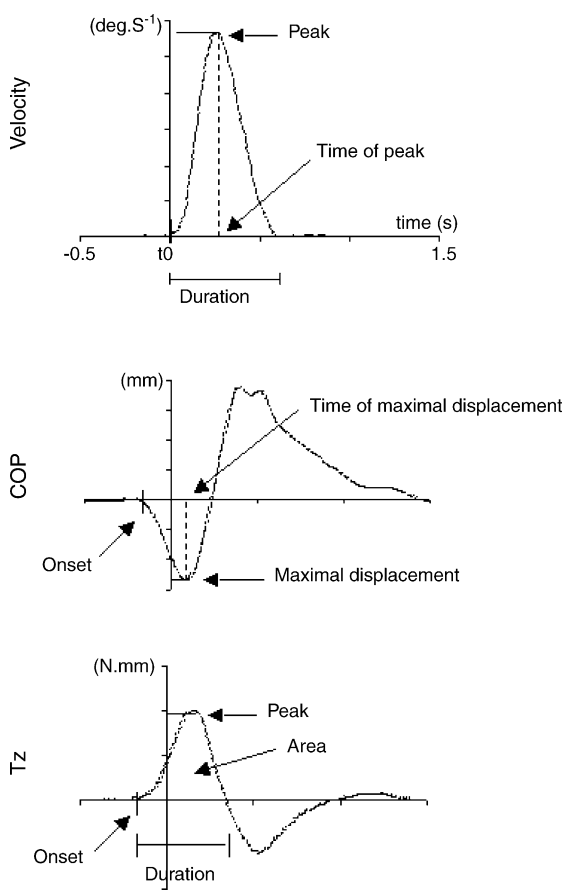

Fig. 1. The mean movement velocity, mean vertical torque (T_z) trace and mean COP anteroposterior displacement for a given subject in the self-paced condition at maximal velocity are displayed from top to bottom. The onset of the angular velocity of the acromion-wrist segment was determined and defined as "time zero" (t_0). We also calculated the magnitude and time of the peak velocity, as well as the movement duration. The COP displacement's onset time and maximal displacement and the time of the maximal backward shift were determined. The two phases of vertical torque were usually clearly recognizable. The onset of T_z was determined, together with the duration, area and peak of the positive phase reflecting the APA. The area of the positive phase occurring before t_0 was also calculated.

The onset of the angular velocity of the acromion-wrist segment was determined and defined as "time zero" (t_0) for arm movement onset. All trials were realigned with respect to t_0 and were analyzed from 0.5 s before to 1.5 s after. We also calculated the magnitude and time of the peak velocity, as well as the movement duration (from t_0 to the return to baseline). With respect to the COP, only its anteroposterior displacement was considered (Fig. 1). The COP displacement's onset time and the magnitude and time of the maximal backward shift were determined. The vertical torque displays two phases: a negative phase (corresponding to a counter-clockwise twist) under all circumstances (i.e. voluntary or passive movement) and a preceding positive phase (corresponding to a clockwise twist) during voluntary movement only (i.e. reflecting postural preparation) [16]. The two phases of vertical torque in this voluntary movement were usually clearly recognizable (Fig. 1). For each trial, the latency of T_z was determined, as well as the duration, area and the maximal magnitude of the positive phase reflecting the APA. The area of the positive phase occurring before t_0 was also calculated. The latency of muscle activation was agreed on by two of the authors after visual inspection of EMG bursts. We determined the onset of the EMG burst relative to its baseline and focused on the ipsilateral biceps femoris, contralateral quadriceps, bilateral tibialis anterior and soleus, which correspond to postural muscles displaying early changes (in contrast to the contralateral biceps femoris and ipsilateral quadriceps). The frequency of muscle activation was also determined and corresponds to the proportion of subjects activating a given muscle.

All latencies were determined with respect to t_0—a negative value means that the event preceded the movement onset.

2.4. Statistical analysis

The mean of 10 trials was calculated in each condition for each individual.

The non-parametric Mann–Whitney test was used to compare movement parameters, T_z, COP and EMG data in each condition (self-paced movement at different velocities, self-paced movement with a load and then externally triggered movement) for young versus elderly subjects. The same test was used for EMG latency when possible, i.e. whenever the frequency of activation was high enough to allow comparison. Hence, we were not able to perform the latter test for the ipsilateral and contralateral soleus muscles in the loaded condition or for the contralateral soleus muscle in self-paced conditions.

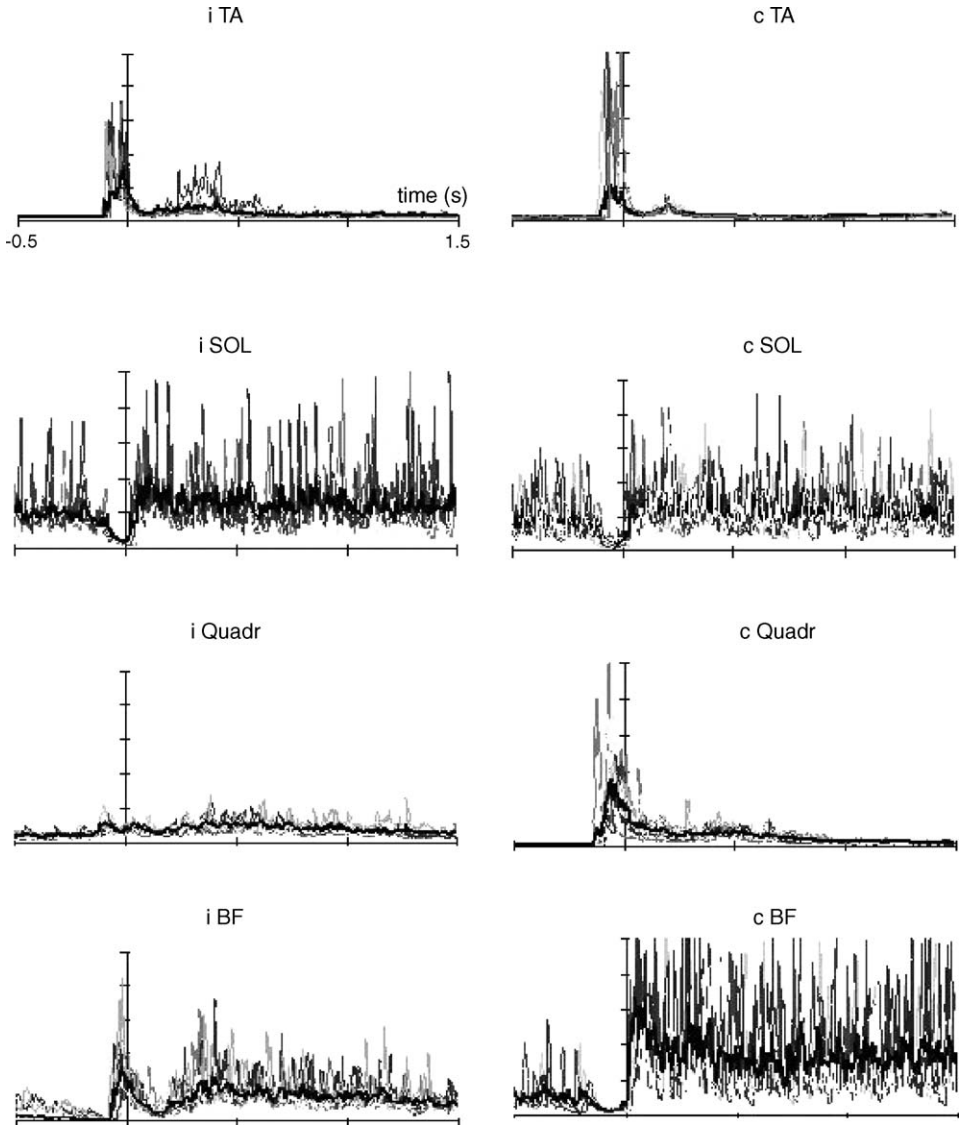

Fig. 2. Typical EMG traces for a representative, elderly subject in the reaction time task condition. EMG activities for 10 trials are superimposed on the mean trace (bold line). All traces are aligned with respect to movement onset (vertical line). TA, tibialis anterior; SOL, soleus; Quadr, quadriceps; BF, biceps femoris; i, ipsilateral; c, contralateral.

Statistical significance was defined as $p < 0.05$ and statistical analysis was performed with SPSS software, Version 9.0.

3. Results

The values indicated in the results correspond to the median (first and third quartiles).

3.1. Self-paced condition

Shoulder flexion was performed at three velocities (low, medium and as quickly as possible), as reflected by the median values of the peak velocity (specified in Table 1).

For slow and medium movements, the movement parameters did not differ between young and elderly subjects. Equally, there were no differences between the two groups in terms of T_z and COP. However, the anticipatory EMG activities we recorded were highly variable and could not usefully be analyzed as in our previous study [17].

Table 1
Median (first and third quartile) velocities in the self-paced condition at different velocities (low, medium and maximal) in young and elderly subjects

Velocities (deg s^{-1}) in self-paced condition	Slow	Medium	Maximal
Young	168 (160; 177)	272 (223; 303)	442 (380; 478)
Elderly	175 (170; 183)	261 (210; 332)	353 (335; 428)

Table 2
Median (first and third quartile) values of characteristic features of T_z and COP during self-paced movement at maximal velocity and in reaction time and loaded conditions in young (Y) and elderly (E) subjects

	Self-paced max			Triggered			Loaded		
	Young	Elderly	*P*	Young	Elderly	*P*	Young	Elderly	*P*
Movement									
Velocity peak (deg s^{-1})	442.4 (379.8; 477.6)	353.3 (334.6; 427.6)		442 (399; 496)	414.1 (362.4; 453)		364.8 (324.1; 387.9)	333.5 (290.3; 359.3)	
Time of peak (s)	0.17 (0.149; 0.222)	0.22 (0.205; 0.238)	*	0.172 (0.155; 0.192)	0.193 (0.153; 0.208)		0.187 (0.164; 0.247)	0.24 (0.202; 0.264)	
Duration (s)	0.388 (0.29; 0.443)	0.386 (0.371; 0.477)		0.349 (0.3; 0.43)	0.359 (0.314; 0.423)		0.404 (0.359; 0.512)	0.443 (0.392; 0.515)	
COP displacement									
Time (s)	−0.256 (−0.378; −0.114)	−0.148 (−0.19; −0.08)	*	−0.09 (−0.15; −0.086)	−0.119 (−0.19; −0.074)		−0.191 (−0.258; −0.16)	−0.157 (−0.256; −0.102)	
Magnitude (mm)	−5.4 (−9.25; −3.8)	−10.3 (−15.3; −4.9)		−9.3 (−20.7; −8.07)	−15.7 (−27.9; −9.5)		−6.8 (−13.5; −4.7)	−11.7 (−15.5; −7.4)	
Time of the maximal displacement (s)	0.112 (0.05; 0.158)	0.154 (0.09; 0.178)		0.128 (0.084; 0.145)	0.124 (0.076; 0.172)		0.074 (−0.001; 0.097)	0.043 (0.005; 0.169)	
T_z (positive phase)									
Onset (s)	−0.177 (−0.223; −0.116)	−0.119 (−0.161; −0.06)	*	−0.07 (−0.086; −0.06)	−0.049 (−0.055; −0.02)	*	−0.194 (−0.21; −0.162)	−0.085 (−0.172; −0.076)	*
Peak (N mm)	7249 (3441; 10713)	7971 (5315; 8865)		7742 (4145; 13447)	11719 (8328; 12818)		8517 (5444; 14910)	11309 (9127; 16739)	
Area prior t_0 (N mm s)	147.2 (90.4; 306.3)	47 (31; 134)	*	42 (23.4; 88.2)	32 (8.2; 33)		199 (179; 507)	98.7 (38.9; 246.5)	
Area (N mm s)	1280 (657; 1844)	1445.5 (887; 1643)		1304 (721; 1952)	1765 (1296; 1968)		1547 (1109; 2721)	2054 (1623; 2575)	
Duration (s)	0.403 (0.333; 0.457)	0.399 (0.361; 0.447)		0.335 (0.288; 0.359)	0.313 (0.281; 0.316)		0.405 (0.372; 0.458)	0.374 (0.336; 0.446)	

Statistically significant differences between the Y and E groups are marked with an asterisk (*).

In rapid, self-paced movement, the time of peak velocity occurred later in elderly subjects. Certain characteristic features of T_z also differed between the two groups (Table 2). This was especially clear in terms of timing: elderly subjects showed a delay in T_z onset ($U = 19$, $p = 0.019$). The area of the positive phase occurring prior to t_0 was clearly smaller ($U = 16$, $p = 0.01$). In contrast, the peak and area of the positive phase of T_z were unchanged. Finally, T_z in self-paced movement at maximal velocity had the same morphology in the two groups but showed decreased anticipation (i.e. delayed onset of T_z and decreased area of the positive phase prior to t_0) in elderly subjects. Concerning the COP, only the displacement onset time was delayed in the elderly group.

There was no significant effect on EMG latencies. A soleus inhibition before movement onset was observed only for the ipsilateral muscle in 7 out of the 10 young subjects (Table 3). This deactivation was also observed in the elderly group for the ipsilateral (8 out of 10 subjects) and contralateral (5 out of 10 subjects) muscles. The first muscle event in young subjects was deactivation of the ipsilateral soleus, followed successively by ipsilateral tibialis anterior and biceps femoris activation. In the elderly group, the first muscle event was activation of the contralateral quadriceps, followed by deactivation of the ipsilateral soleus.

3.2. Reaction time condition

Kinematic parameters (reaction time, peak velocity and time of peak) did not differ between the two groups (Table 2).

T_z was delayed in elderly subjects ($U = 9.5$, $p = 0.002$). Neither the other T_z parameters nor the COP parameters differed between the two age groups (Table 2).

The ipsilateral BF was activated more often in elderly subjects. The contralateral quadriceps was activated in all elderly subjects, and inhibition of both soleus muscles was also more frequent in this group (5 out of 10 young subjects and 8 out of 10 elderly subjects for the ipsilateral soleus). In both groups, the first muscle event was deactivation of soleus muscle.

3.3. Loaded condition

Kinematic parameters were similar in the two groups.

The onset of T_z was delayed in the elderly group ($U = 23$, $p = 0.041$), whereas the other T_z characteristics did not differ between the groups. Equally, no difference for COP was observed between the two groups.

The ipsilateral BF was activated in all elderly subjects. Activation of the ipsilateral TA was significantly delayed in the elderly group. Young subjects did not show visible soleus deactivation, whereas this phenomenon was observed in 9 out of 10 elderly subjects for the ipsilateral soleus and in 5 out of 10 for the contralateral soleus. The first muscle event

Table 3
Median (first and third quartile) EMG latencies for 10 elderly subjects and 10 young subjects

	Young			Elderly		
	Trig	Self max	Load	Trig	Self max	Load
I BF	−0.1 (−0.132;−0.076) 7/10	−0.152 (−0.16; −0.114) 7/10	−0.175 (−0.198; −0.154) 8/10	−0.096 (−0.104; −0.072) 9/10	−0.13 (−0.146;−0.119) 8/10	−0.134 (−0.166; −0.112) 10/10
C Quadr	−0.104 (−0.14;−0.062) 6/10	−0.088 (−0.126; −0.078) 3/10	−0.184 (−0.192; −0.128) 5/10	−0.114 (−0.129; −0.101) 10/10	−0.176 (−0.204; −0.147) 4/10	−0.144 (−0.181; −0.14) 6/10
I TA	−0.132 (−0.16;−0.088) 9/10	−0.174 (−0.204; −0.141) 6/10	−0.216 (−0.228; −0.192) 9/10	−0.12 (−0.148; −0.092) 9/10	−0.116 (−0.192; −0.04) 5/10	−0.088 (−0.128; −0.066) 7/10
C TA	−0.132 (−0.176; −0.099) 8/10	−0.128 (−0.147; −0.112) 4/10	−0.162 (−0.17; −0.142) 6/10	−0.112 (−0.148; −0.092) 9/10	−0.148 (−0.196; −0.12) 5/10	−0.1 (−0.138;−0.028) 7/10
I Sol	−0.128 (−0.194; −0.11) 5/10	−0.212 (−0.216; −0.160) 7/10	0/10	−0.124 (−0.161; −0.103) 8/10	−0.158 (−0.185; −0.114) 8/10	−0.140 (−0.190; −0.1) 9/10
C Sol	−0.152 (−0.178; −0.048) 5/10	0/10	0/10	−0.112 (−0.138; −0.094) 6/10	−0.168 (−0.194; −0.086) 5/10	−0.112 (−0.206; −0.070) 5/10

The number of subjects (X) activating a given muscle is shown as (X/10).

corresponded to activation of the ipsilateral TA in young subjects and activation of the contralateral quadriceps in the elderly group.

4. Discussion

The postural preparation associated with unilateral arm voluntary movement was analyzed using COP displacement and EMG data, together with a rarely exploited biomechanical parameter—vertical torque. At low movement velocity, elderly subjects did not show any impairments in stability. At maximal velocity, the vertical torque was delayed in the elderly group. Despite this decrease in anticipation, the movement was performed at the same velocity as in the younger group. Analysis of the EMG data showed that elderly subjects adopted different muscle strategies (such as a modification of the activation sequence and increased activation of the thigh muscle) in order to perform the same movement with less stability.

An asymmetric movement (such as right arm raising) causes a counter-clockwise twist (as viewed from above) applied to the base support [16]. This moment has repercussions (because of the action and reaction principle) on all body segments and, at the end of the chain, causes a modification in the reaction exerted by the subject on the support. This phenomenon results in negative deviation of T_z, corresponding to segmental displacement. A positive phase precedes the negative phase only when the movement is performed actively. Consequently, this positive phase reflects the postural preparation associated with voluntary movement and corresponds to a clockwise torsion of the body (as viewed from above).

APAs revealed by T_z allow the body to rotate in the opposite direction to that induced by the arm movement. A counter-clockwise rotation should occur, due to reactive forces and moments produced by the arm movement. The positive phase of T_z before t_0 and during the first phase of movement execution is always in the direction opposing the trunk rotation caused by arm movement. Hence, APAs revealed by T_z are generated in order to locally stabilize joints that are affected by arm movements and thus assist movement [17].

T_z is the overall consequence of local postural activities (such as muscle activity, joint movements and visco-elastic properties) and ultimately provides information on all the anticipatory phenomena associated with voluntary movement. Analysis of T_z must be combined with that of EMG, which gives information on various balance strategies.

We observed a delay in rotational postural preparation (T_z) in the elderly group compared with young subjects, whatever the movement condition, i.e. self-paced movement with and without load or triggered movement. In contrast, anteroposterior postural preparation (COP) was only delayed in elderly subjects in the self-paced condition without load. This age-related decrease in anticipation was ot compensated by more intense postural preparation during the first phase of movement execution and had no consequences on voluntary movement execution. Nevertheless, other studies with similar tasks have shown that in order to counteract decreased anticipation, the time delay between postural and voluntary muscle activation increased when the biomechanical requirements of the intentional movement increased [19], and that voluntary movement was only initiated after postural stabilization had occurred. Thus, local adjustment strategies (revealed by EMG activity) represent the only way that the elderly can perform the same movement at the same velocity, despite their lower stability.

In the self-paced condition at maximal velocity with and without load, the sequence of muscle activation in young subjects was distal to proximal. In agreement with the literature, this sequence tended to be disrupted in elderly subjects [11]. For the latter, the first muscle event was activation of the contralateral quadriceps. Moreover, for young subjects in the condition without load, soleus inhibition was only present for the ipsilateral muscle, whereas inhibition was present for both muscles (i.e. ipsilateral and contralateral) in elderly subjects. In agreement with the literature [20], this asymmetric, anticipatory sequence was specific for the forthcoming (unilateral) movement in young subjects but not in elderly subjects. The elderly subjects performed the rapid movement with less stability, as reported by Mankowski et al. [21]. Thus, a hip strategy was used to maintain stability when the inertia of movement increased, as reported in the literature [22,23]: the thigh muscles were activated before those of the leg.

In the triggered condition, a different strategy was revealed. The sequence of muscle activation was distal to proximal in both groups and the strategy corresponded to more frequent activation of thigh muscles in the elderly group, as previously reported by Woollacott and Manchester [14]. This observation suggests a change in strategy in order to compensate for the lack of stability induced by decreased postural preparation. Thus, in the triggered condition, the strategy corresponded to an increase in activation of certain muscles rather than a sequence modification.

On the other hand, we noticed that the velocity in the self-paced condition without load was not truly maximal, compared with the speed observed in the triggered condition. Elderly subjects performed the task more cautiously, with a wider safety margin. Blaszczyk et al. drew the same conclusion with another movement paradigm [10] when studying postural destabilization in response to cyclic pull-and-push tasks in young and elderly subjects, performed on both rigid and foam surfaces. They observed a reduction in arm movement velocity in young subjects during the fast movement condition performed on a foam surface. This effect was not observed in the elderly group, even though the task performed on the foam surface was more destabilizing.

We note that at low velocity, elderly subjects have no impairment in stability and can then perform the movement with suitable postural preparation, just like young subjects. Our results are similar to those of Mankowski et al., who studied a leg lift with knee flexion at low speed and as quickly as possible in young and elderly subjects [21]. These authors reported that at low velocity, elderly subjects performed the movement with the appropriate anticipation.

On the other hand, we cannot exclude an influence of arm flexion with a final prehension movement on APAs, relative to simple arm-raising. Compared to a classic arm-raising movement, our paradigm can be considered as less destabilizing. The contact with the handle may reassure subjects and may require less APA.

In the present study, vertical torque data showed that postural preparation associated with fast, voluntary arm movement was delayed in elderly subjects. Lower stability was compensated for by different muscle strategies in order to maintain balance. The technique described here can potentially be extended to the study of APAs associated with unilateral arm-raising movement in patients with Parkinson's disease or cerebellar disease. We intend to determine whether or not biomechanical parameters and the characteristics of voluntary movement are modified in patients, in comparison with age-matched, healthy controls.

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